WORKED SOLUTION

STEP II 2003, Question 10

Pure • Acceleration

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q10

The key idea

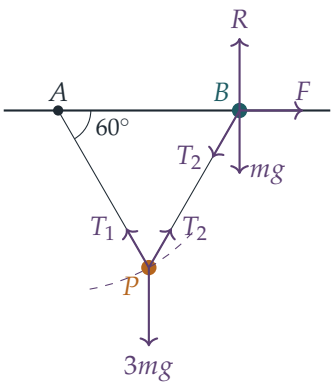
At the start, $AP = PB = AB = \ell$, so APB is an equilateral triangle. Because AP stays taut, P moves on a circle about A. Everything begins at rest, so there is no centripetal term, and P's initial acceleration must be along the tangent to that circle. This fixes its direction. Then P is always at the apex of an isosceles triangle on AB, so P's horizontal position is always half of B's. For the final part, write down Newton's second law for B and for P. Resolving for P along its direction of acceleration (perpendicular to AP) means the tension in AP never appears.

Solution

We model B and P as particles and the string as light and inextensible. Both halves of the string stay taut. Take A as the origin, with the wire along the x-axis and B initially at $(\ell, 0)$. Let θ be the angle PAB. Since $AP = PB = \ell$,

$$B = (2\ell\cos\theta,\ 0), \qquad P = (\ell\cos\theta,\ -\ell\sin\theta),$$

and at the start $\theta = \frac{\pi}{3}$.



The direction of P's acceleration

Since $AP = \ell$ is fixed, P moves on a circle of radius ℓ centred at A (dashed in the diagram). The acceleration of a point on a circle has a component $\ell\dot{\theta}^2$ towards the centre and a component $\ell\ddot{\theta}$ along the tangent. At the start $\dot{\theta} = 0$, because the system is released from rest. So the initial acceleration of P points along the tangent, which is perpendicular to AP.

AP makes an angle of 60° with the horizontal, so the perpendicular to AP makes an angle of 30° with the horizontal. If the acceleration has magnitude a, then

$$a_1 = a\cos 30^\circ = \tfrac{\sqrt{3}}{2}a, \qquad a_2 = a\sin 30^\circ = \tfrac{1}{2}a,$$

and so $a_1 = \sqrt{3}\,a_2$, as required.

In terms of θ: differentiating P's position and setting $\dot{\theta} = 0$ gives an initial acceleration of $\ell\ddot{\theta}(-\sin\theta, -\cos\theta)$. With $\ddot{\theta} > 0$, P moves down and towards A. Its magnitude is $a = \ell\ddot{\theta}$.

The acceleration of B

The x-coordinate of B is always exactly twice that of P, because P lies on the perpendicular bisector of AB. Differentiating twice shows that B's acceleration is always twice the horizontal acceleration of P. So the initial acceleration of B has magnitude $2a_1$, directed towards A, as required.

> **Ask yourself** Why can we say B moves towards A, and which way does friction act?

As P falls, θ increases, so $x_B = 2\ell\cos\theta$ decreases. So B slides towards A, and friction on B acts away from A (in the positive x-direction). Here we are assuming that B really does move. The question confirms this by giving the friction as $\frac{\sqrt{3}}{6}R$, which is its value during sliding.

The magnitude of the initial acceleration

Let T_1 and T_2 be the tensions in AP and PB. Let R be the normal reaction on B and $F = \frac{\sqrt{3}}{6}R$ the friction. Write a for the magnitude of P's initial acceleration, so $a_1 = \frac{\sqrt{3}}{2}a$, and B's acceleration is $2a_1 = \sqrt{3}\,a$ towards A.

Bead B. The string BP pulls B at 60° below the horizontal, towards A. B has no vertical acceleration, so resolving vertically gives

$$R = mg + \tfrac{\sqrt{3}}{2}T_2.$$

Resolving horizontally, taking the direction towards A as positive:

$$m(\sqrt{3}\,a) = \tfrac{1}{2}T_2 - \tfrac{\sqrt{3}}{6}\left(mg + \tfrac{\sqrt{3}}{2}T_2\right) = \tfrac{1}{4}T_2 - \tfrac{\sqrt{3}}{6}mg,$$

so

$$T_2 = 4\sqrt{3}\,ma + \tfrac{2\sqrt{3}}{3}mg. \tag{1}$$

Particle P. Resolve along the direction of P's acceleration, that is, down and towards A at 30° below the horizontal. This direction is perpendicular to AP, so T_1 contributes nothing. The weight $3mg$ makes an angle of 60° with this direction, so it contributes $3mg\cos 60° = \frac{3}{2}mg$. The tension T_2 acts along PB, up and away from A at 60° above the horizontal. That makes an angle of 150° with the direction of acceleration, so it contributes $T_2\cos 150° = -\frac{\sqrt{3}}{2}T_2$. Hence

$$3ma = \tfrac{3}{2}mg - \tfrac{\sqrt{3}}{2}T_2. \tag{2}$$

Substituting (1) into (2):

$$3ma = \tfrac{3}{2}mg - \tfrac{\sqrt{3}}{2}\left(4\sqrt{3}\,ma + \tfrac{2\sqrt{3}}{3}mg\right) = \tfrac{3}{2}mg - 6ma - mg,$$
$$9ma = \tfrac{1}{2}mg,$$

so the magnitude of the initial acceleration of P is

$$a = \frac{g}{18},$$

as required.

The answer is consistent: (1) gives $T_2 = \frac{8\sqrt{3}}{9}mg > 0$, so the string PB is taut. Then $R = \frac{7}{3}mg > 0$. Resolving horizontally for P gives $T_1 = T_2 + 6ma_1 = \frac{19\sqrt{3}}{18}mg > 0$, so AP is taut too.

Checks

- Vertical equation for P, which we did not use: $\frac{\sqrt{3}}{2}(T_1 + T_2) - 3mg = \frac{35}{12}mg - 3mg = -\frac{1}{12}mg$. This equals $-3ma_2 = -3m \cdot \frac{g}{36}$, as it should.
- Numerically, we differentiated the positions of B and P to get $a_1/a_2 \approx 1.7320508$ and $a_B/a_1 = 2$. We then solved the three force equations for T_1, T_2 and $\ddot{\theta}$ with $g = 9.81$. This gave $|a| = 0.545 = g/18$, with $T_2/mg \approx 1.5396$ and $R/mg \approx 2.3333$.
- Without friction, (1) would be $T_2 = 2\sqrt{3}\,ma$ and (2) would give $6ma = \frac{3}{2}mg$, so $a = g/4$. Friction should slow things down, and $g/18 < g/4$.

Common slips

- Including a centripetal term. At the instant of release $\dot{\theta} = 0$, so the acceleration is purely tangential.
- Forgetting that T_2 pulls B downwards, which increases R and therefore the friction. Taking $R = mg$ gives the wrong answer.
- Resolving for P horizontally and vertically. This works, but it brings in T_1 and needs one more equation. Resolving perpendicular to AP removes it.

Beyond STEP

This is a constrained system with one degree of freedom: everything is fixed by the single angle θ. Writing positions in terms of θ and differentiating is the first step of Lagrangian mechanics. In Lagrangian mechanics, constraint forces such as T_1 are removed systematically rather than by a clever choice of direction. Friction does not fit neatly into that framework, which is one reason questions like this are good practice in Newtonian methods. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers constrained motion and non-uniform circular motion carefully.

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